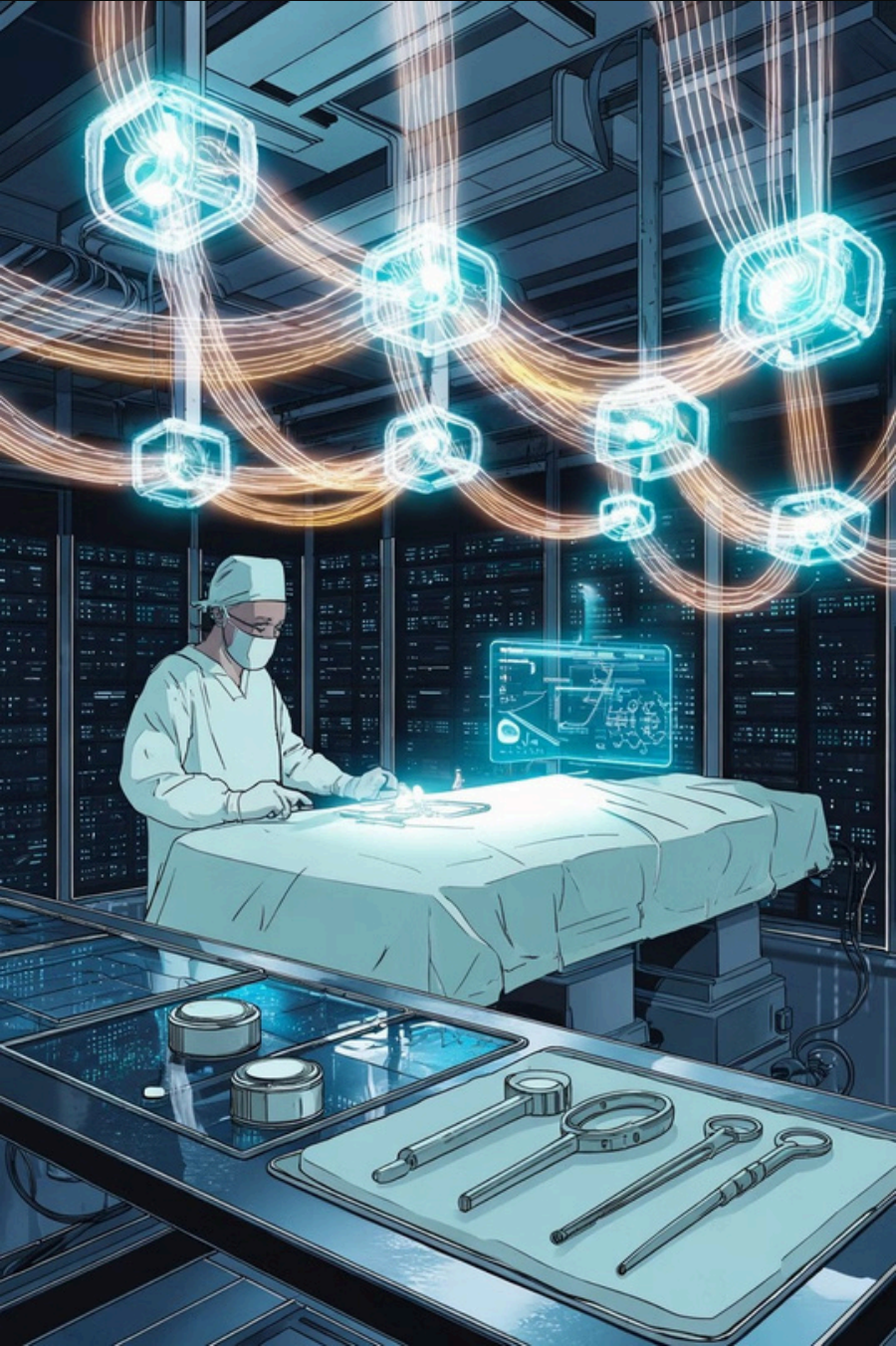




Decentralized Blood Donor Application

A Secure and Transparent Platform Using Blockchain

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Topics to be Covered

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Abstract

Blood shortage is one of the major causes of preventable deaths worldwide, affecting emergency surgeries, trauma care, and chronic illness treatment.

This project presents BloodLink — a Decentralized Blood Donor Application built on the Ethereum Sepolia Blockchain using SHA-256 hash verification and smart contracts.

The system uses a geo-based matching algorithm with MongoDB 2dsphere indexing to connect compatible donors and recipients within a 50 km radius based on blood group and location.

Donors are instantly rewarded with Blood Donor Coins (BDC) and NFT Donation Certificates on every successfully confirmed donation.

The platform supports hospitals, blood banks, and veterinary care institutions through the Dogin animal donation module.

The AI Analyst module powered by Groq + LLaMA 3.3-70B provides real-time platform health reports and shortage alerts.

BloodLink reduces blood shortages caused by poor donor engagement and improves healthcare equity through a community-driven, blockchain-powered ecosystem.

Introduction

The Crisis

Blood availability is a life-critical resource severely limited by **poor donor engagement**, lack of transparency, and outdated centralized systems. According to healthcare studies, a significant percentage of emergency deaths occur due to **unavailability of compatible blood**, especially during disasters and in remote regions.

Why Traditional Systems Fail

Traditional systems — manual donor registries, camp drives, and centralized blood bank databases — fail to address the root cause of donor disengagement. Centralized systems are vulnerable to **data breaches**, lack cross-institutional visibility, and offer donors **no transparency** into how their blood is used.

The Opportunity

With advancements in blockchain technology, DApps, and token-based incentive systems, it is now possible to build platforms that continuously engage donors and track every donated blood unit transparently.

BloodLink Solution

- Ethereum blockchain for immutable donation records
- Geo-based matching for real-time donor-recipient connection
- BDC token rewards and NFT certificates for donor motivation
- AI-powered analytics for continuous platform monitoring

Problem Statement

Donor Disengagement

Traditional systems have no mechanism to continuously motivate or reward donors, leading to critically low repeat donation rates.

Centralized Vulnerability

Blood bank records stored in centralized databases are prone to data breaches, unauthorized modifications, and fraud.

No Real-Time Matching

Existing systems do not support geo-based real-time matching between donors and recipients, causing dangerous delays during emergencies.

Lack of Transparency

Donors have no visibility into how or where their donated blood is used, reducing trust and long-term motivation.

Blood Wastage

Poor inventory management and expiry tracking lead to significant blood wastage across hospital networks.

Data Silos

Hospitals and blood banks operate independently with no unified platform for cross-institutional coordination and visibility.

Objective

1

Real-Time Management

Develop a real-time blood donation management system that continuously monitors blood availability and connects donors with hospitals in need.

2

Blockchain Verification

Integrate blockchain-based verification and geospatial donor matching for accurate and transparent donor-recipient coordination.

3

Minimize Wastage

Minimize blood wastage and fulfillment delays using inventory management, geo-matching, and automated expiry handling.

4

User-Friendly Platform

Design a non-invasive, user-friendly platform that encourages regular donor participation without disrupting daily activities.

5

Incentive Mechanism

Implement a reliable incentive mechanism using Blood Donor Coins (BDC) and NFT Donation Certificates, rewarded instantly on confirmed donation.

6

Scalable Architecture

Build a scalable, extensible architecture supporting cross-chain blockchain, government API connectivity (eRaktaKosh), and mobile development.

Existing System — Drawbacks



Centralized Databases

Managed independently by individual hospitals or government agencies — vulnerable to tampering, fraud, and unauthorized modifications.



Manual Registries

Camp-based drives do not engage donors continuously or provide real-time matching — prone to error, delay, and record manipulation.



No Incentive Mechanism

Donors receive no tangible reward or recognition for their life-saving contribution, reducing long-term platform engagement.



Data Silos

Hospitals operate in isolated data silos — no cross-institutional visibility or blood sharing coordination between facilities.



No Geo-Matching

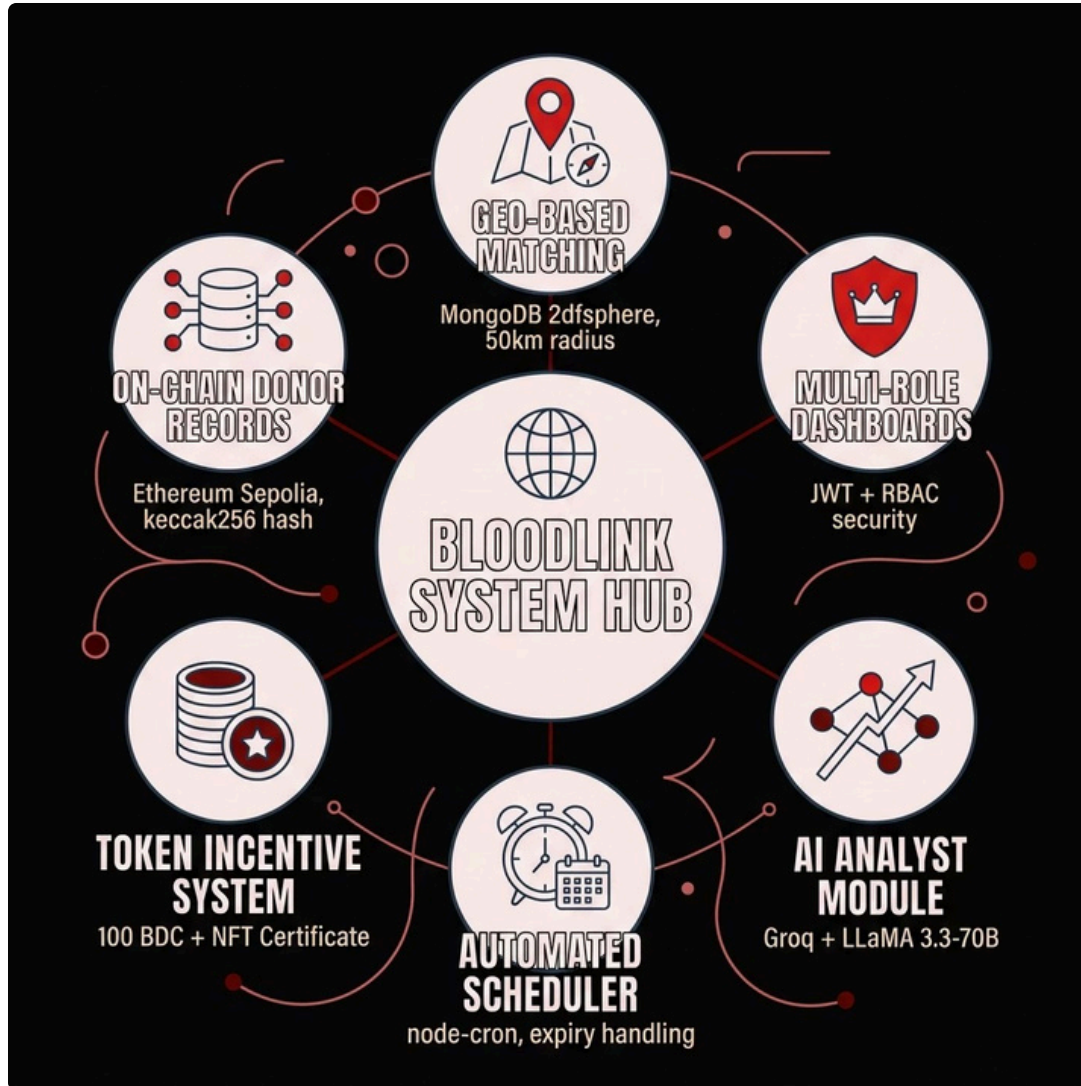
Compatible donors and recipients cannot be connected instantly during emergencies — no real-time geospatial matching capability.



Zero Transparency

Donors have zero visibility into how or where their donated blood is used after donation — eroding trust and motivation.

Proposed System — BloodLink



Key Innovations

On-Chain Records

Donor profiles and hospital blood requests stored immutably on Ethereum Sepolia blockchain — tamper-proof and permanently verifiable.

Automated Notifications

Email alerts via Nodemailer+Gmail SMTP and real-time dashboard updates triggered at every donation lifecycle stage.

Inter-Hospital Transfer

Hospitals can source blood from nearby facilities with surplus stock through the transfer request module.

BloodLink System Modules

BloodLink is structured around six core modules, each designed to manage a specific aspect of the blood donation ecosystem, ensuring efficiency, transparency, and engagement.



Donor Module

Facilitates donor registration, allows viewing and accepting blood requests via an interactive map, and rewards completed donations with BDC tokens and NFT certificates.



Admin Module

Provides platform-wide user management, dispute resolution, comprehensive monitoring of donations and the BDC token economy, and generates AI-powered health reports.



Blockchain Module

Records donation hashes on the Ethereum Sepolia testnet via MetaMask, mints NFT certificates, and instantly awards BDC tokens upon successful donation completion.



Hospital Module

Enables hospitals to post urgent blood requests, confirm donation receipts, and manage real-time inventory for all blood groups to prevent shortages and wastage.



AI Analyst Module

Offers role-aware AI chat with live database context, generating insightful reports and supporting administrators in executing critical actions and alerts.

Methodology

The BloodLink system employs a robust, multi-modal methodology that integrates two independent verification streams to ensure precise, secure, and transparent donor matching.

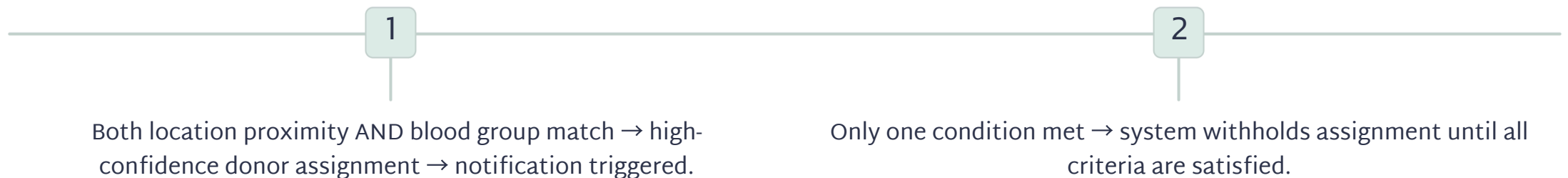
Stream 1 — Geospatial Donor Matching

- Donor registers with GPS coordinates and blood group.
- Hospital posts blood request with location and urgency.
- System queries MongoDB 2dsphere index for donors within 50 km.
- Filters applied: blood group compatibility + 56-day donation cooldown.

Stream 2 — Blockchain-Based Verification

- SHA-256 donation hash computed from: Donor ID + Hospital ID + Request ID + Timestamp.
- Hash recorded on Ethereum Sepolia testnet on donation completion.
- System verifies stored hash against on-chain record to detect tampering.
- BDC tokens awarded only after on-chain hash verification passes.

Decision Logic (Fusion)



RESULTS & DISCUSSION

Our evaluation demonstrates BloodLink's strong performance across critical functions, alongside identifying key areas for future enhancement.



Rapid Donor Matching

Successfully identifies compatible donors within seconds, significantly reducing response times.



Precise Geo-Matching

Accurately pinpoints donors within a 50 km radius, optimizing proximity-based assignments.



Robust Verification

Strictly enforces donation cooldowns and blockchain verification for tamper-proof records.



Timely Notifications

Email, dashboard, and BDC token alerts triggered instantly upon donation completion.



Effective AI Analysis

AI analyzer consistently generated correct role-specific reports and critical shortage alerts.

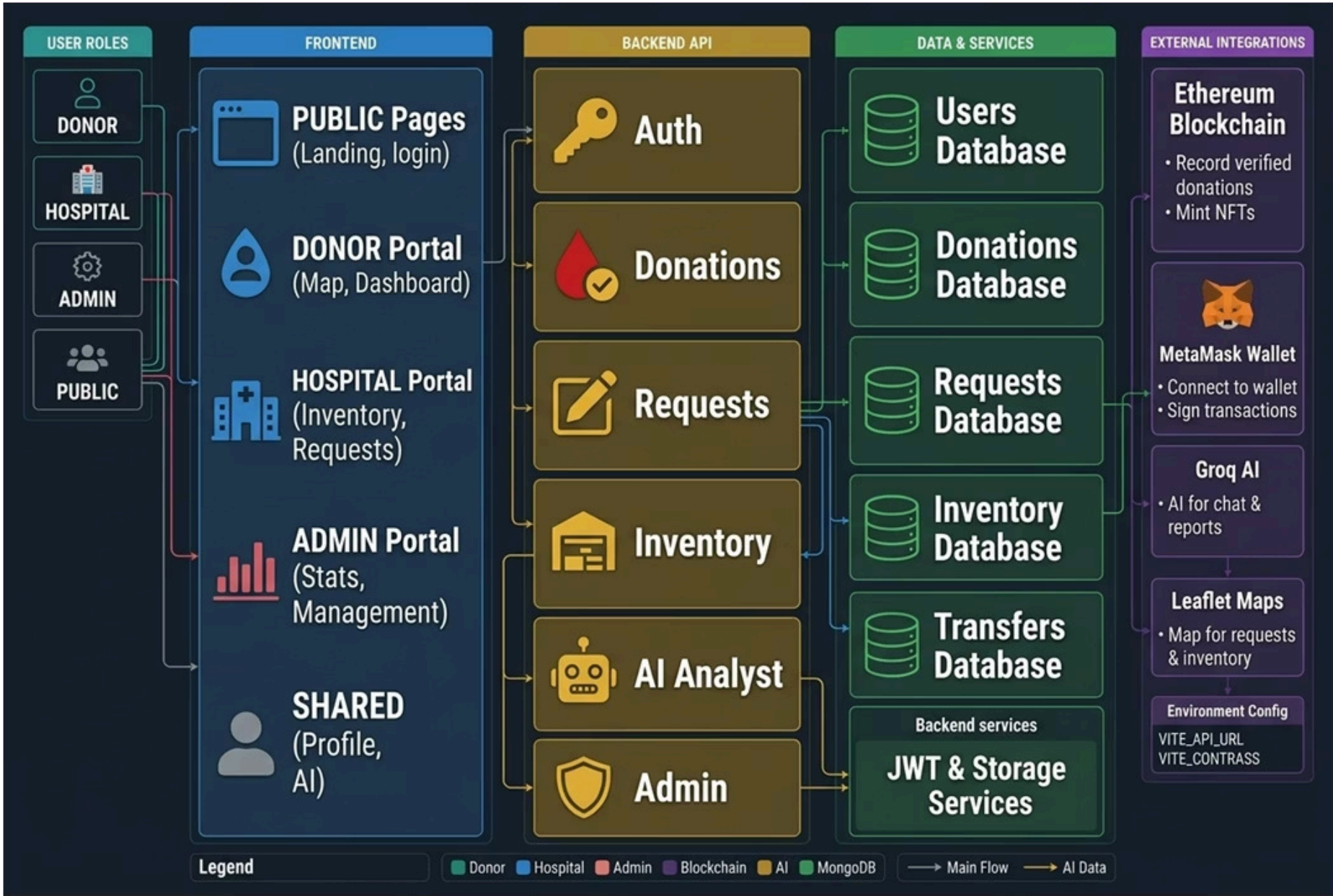
Limitations Identified



Imprecise GPS: Current GPS accuracy can occasionally reduce matching precision, particularly in dense urban environments.

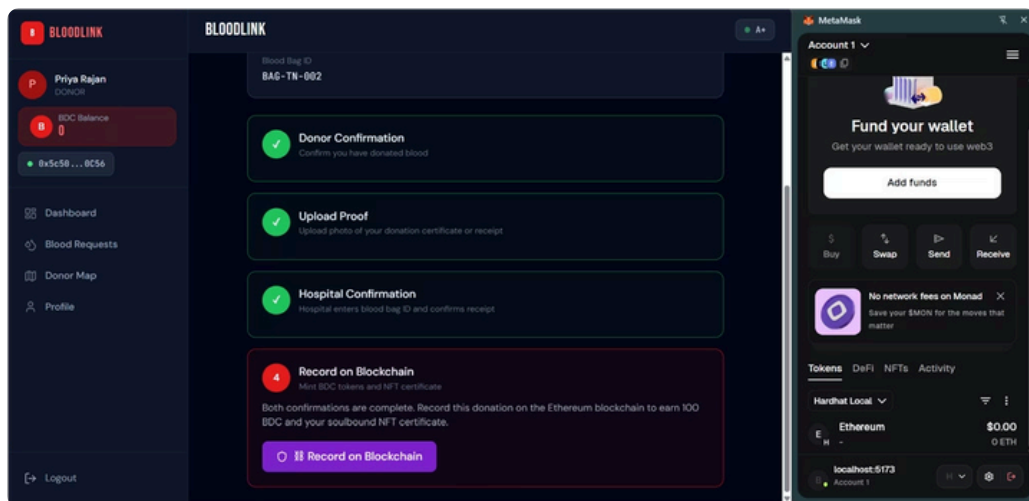
Sepolia Testnet Latency: Dependence on the Sepolia testnet leads to occasional blockchain transaction delays, impacting real-time updates.

UML Diagram



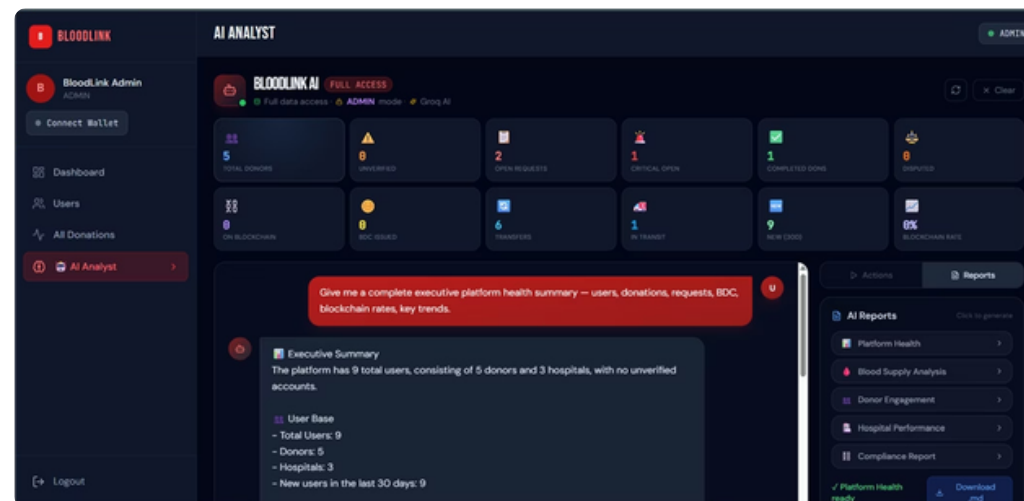
BloodLink Platform Screenshots

These screenshots highlight key user interfaces of the BloodLink platform, demonstrating the donor workflow and admin analytics capabilities.



Donor Portal

Donation confirmation flow with blockchain recording and wallet integration.



AI Analyst Dashboard

Administrative analytics and platform monitoring interface.

Conclusion

The proposed BloodLink Decentralized Blood Donor Application successfully demonstrates an effective and practical approach to managing blood availability in real time. By integrating geo-based donor matching with Ethereum blockchain verification, the system overcomes the core limitations of traditional centralized blood bank systems. The multi-modal approach—combining location proximity, blood group compatibility, and 56-day cooldown validation—ensures accurate, transparent, and timely donor-request matching with minimal mismatches. The BDC token reward and NFT certificate mechanism enhances donor engagement, builds trust, and drives long-term repeat participation.

The AI Analyst module powered by Groq + LLaMA 3.3-70B enables administrators and hospitals to make faster, more informed decisions without manual data analysis. The system is cost-effective, non-invasive, and scalable for deployment across small clinics, large hospital networks, and veterinary care institutions. With further improvements and real-world deployment, BloodLink has the potential to significantly reduce preventable blood shortage deaths and contribute to a safer, more accessible blood donation ecosystem for both human and veterinary healthcare.